Programming 1 | Lab Week 3

Conditions and Loops

Conditions and Loops

- if
- elif
- else
- While True
- For i in range(8)

Live demo in Visual Studio Code

Your tasks | 45 minutes

1. Create multiple falling obstacles using a **for loop** and a **list** (together)
2. Prevent the player from moving off the left side of the screen using an **if** condition
3. If an obstacle goes under the screen reset that obstacle to the top
4. If the player touches an obstacle, deduct health by 20 and reset that obstacle to the top
5. If health reaches 0 print a message and exit the game directly using **pygame.quit()** and **sys.exit().**

Challenges | For faster students

1. The player can currently run off the right side of the screen. Create a Finish Line the Player can reach. *Hint: Remember that the player's width is tied to player_hp*

2. Currently, every obstacle deals 20 damage. Change this so that the obstacles deal damage equal to their size. Use the obstacle's height (**obs.height**) to deduct the exact same amount of HP from the player upon collision

3. Make the game harder over time. Replace the fixed falling speed with a new variable **fall_speed** = 4.0. Every time an obstacle resets to the top (meaning the player successfully dodged it), increase **fall_speed** by 0.1

Compare your solutions

How can these ideas be applied in the real world?

Solution

The solution will be uploaded to the ILIAS folder.

Please take a look before the next exercise.